

DATA ARCHITECTURE

What

The Automated Network Request Management solution uses a structured ServiceNow data model to store and manage information required throughout the network-request lifecycle.

A dedicated custom table is created to maintain network-related request information. The table contains fields for request identification, assignment, requester details, payment information, connection type, address, and approval status.

The data model provides a structured foundation for forms, approvals, automation, operational tracking, and reporting.

Why

A well-defined data architecture is required to ensure that information entered by users is:

- Stored consistently.
- Easy to retrieve and track.
- Available to approval and fulfilment processes.
- Suitable for automation.
- Accurate enough for reporting and auditing.
- Controlled through ServiceNow access and platform capabilities.

Using a dedicated table also provides a clear separation between the application's network-specific information and unrelated platform data.

How

The data architecture will be implemented in two activities:

1. Creation of the custom table
2. Creation and configuration of fields

Activity 1 – Creation of Table

What

Create a dedicated custom ServiceNow table to store the required network database/request information.

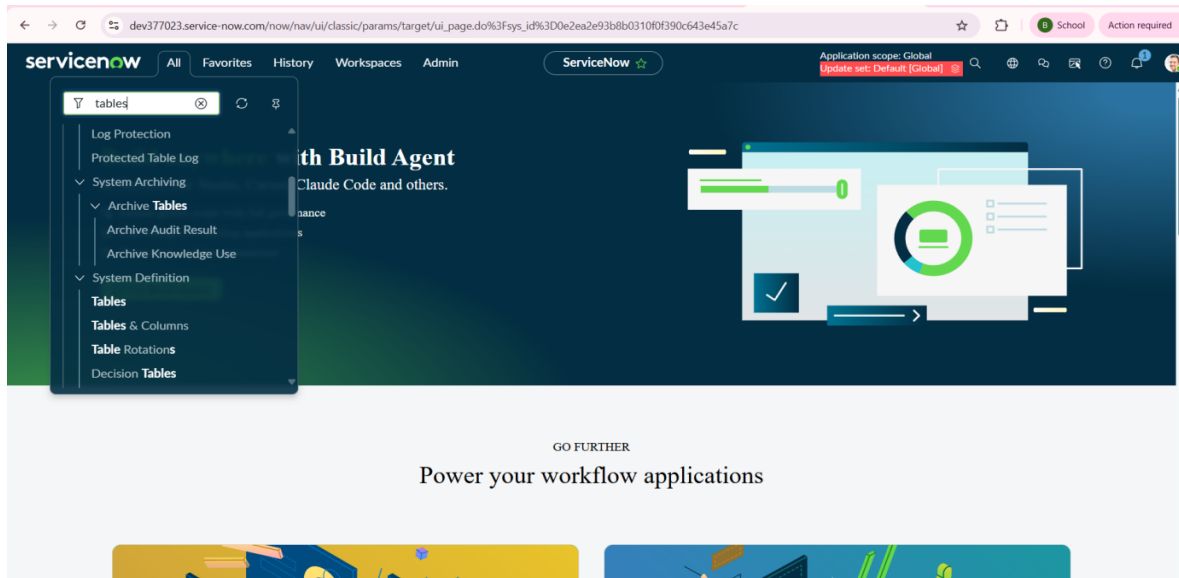
Why

The table acts as the primary data structure for storing records generated by the network request process. Creating a dedicated table makes the data easier to manage, automate, and report on.

How

1. Navigate to:

All → System Definition → Tables

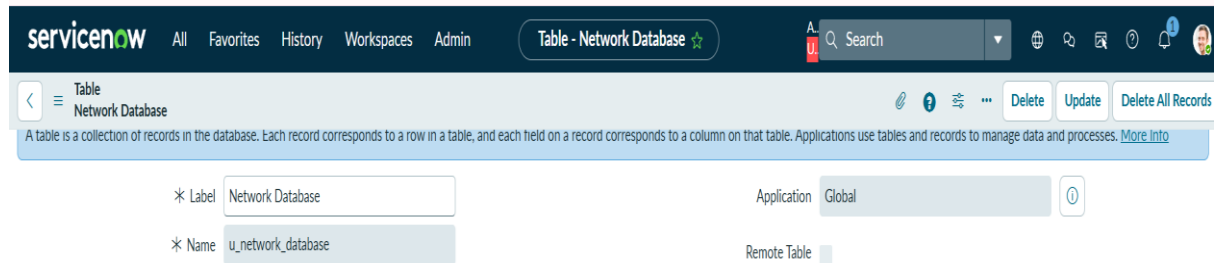


2. Click **New**.
3. Enter the table information provided in the project requirements:
Label: Database Tables
Name: u_database_table
4. Review the configuration.
5. Click **Submit**.
6. Search for **Database Tables** in the table list.
7. Open the newly created table to verify that it exists.

	Label	Name	Extends table	Extensible	Updated
☐	Network Database	u_network_database	(empty)	false	2026-09-01 21:07:12
	Collection Item M2M	sn_udc_collection_item_m2m	(empty)	false	2026-08-31 17:08:12
	Store App Write Audit ACL	store_app_write_audit_acl	(empty)	false	2026-08-31 15:16:18
	Off-Glide Routing Context	sys_og_routing_context	(empty)	false	2026-08-31 15:08:39
	Data Snapshots Unsupported Reason	pa_cdc_unsupported_reasons	(empty)	false	2026-08-31 15:08:36
	Indicators to Data Snapshots Mapping	pa_cdc_indicator_analytics_mapping	(empty)	false	2026-08-31 15:08:35
	Indicator Source Table Row Count	pa_cdc_indicator_table_row_count	(empty)	false	2026-08-31 15:08:34
	Data Snapshots exclusion	pa_cdc_exclusions	(empty)	false	2026-08-31 15:08:34
☒	PA Data Snapshot Usage	pa_data_snapshot_usage	Platform Analytics Event Aggregate	false	2026-08-31 15:08:33
	PA KPI Configuration	pa_dm_kpi_configuration	Application File	false	2026-08-31 15:08:32
	Data Snapshot Indicator	pa_data_snapshot_indicator	PA Indicators Definition	false	2026-08-31 15:08:31
	Data Snapshots Hierarchy	pa_cdc_hierarchy	(empty)	false	2026-08-31 15:08:29
	Data Snapshots source to CDC Table Mapping	pa_cdc_fact_mapping	(empty)	false	2026-08-31 15:08:29
	PA Cache Filter	pa_dm_cache_filter	(empty)	false	2026-08-31 15:08:28
	NAM Supported Fields	pa_cdc_supported_fields	(empty)	false	2026-08-31 15:08:28
	Data Snapshots source Replicate till	pa_cdc_source_replicate_till	(empty)	false	2026-08-31 15:08:27
	DB View ID Mapping	pa_dm_cdc_dbview_id_mapping	(empty)	false	2026-08-31 15:08:26

Capture the **Table** configuration form showing:

- Label: **Database Tables**
- Name: **u_database_table**



The screenshot shows the ServiceNow interface for configuring a table. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', 'Admin', and a 'Table - Network Database' button. A search bar and user profile are on the right. Below the navigation bar, the table name 'Network Database' is displayed. A descriptive text box states: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)'. The configuration form includes fields for 'Label' (set to 'Network Database'), 'Name' (set to 'u_network_database'), 'Application' (set to 'Global'), and 'Remote Table' (empty).

Outcome

The custom **Database Tables (u_database_table)** table is successfully created and is ready for field configuration.

Activity 2 – Creation of Fields

What

Configure the fields required to capture and manage network database/request information.

Why

The fields define the information that users and fulfilment teams need to provide or maintain. Appropriate field types improve data consistency, while reference fields allow records to be associated with existing ServiceNow users.

How

1. Navigate to:
All → System Definition → Tables
2. Search for **Database Tables**.
3. Open the **Database Tables** table.
4. Use the table's column/field configuration to create the required fields.
5. Configure the fields according to the following specification.

Required Fields

Field	Type	Reference
Database Number	String / Auto Number	—
Assignment to	Reference	User
Mobile Number	String	—
Total Amount	String	—
Mode of Payment	String	—
Type of Connection	String	—
Customer Address	String	—
Approval Status	String	—
Requested For	String	—

Column label	Type	Reference	Max length	Default value	Display
Request For	String	(empty)	40		false
Request Number	String	(empty)	10		false
Date of Enquiry	Date	(empty)	40		false
Customer Address	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Assigned To	Reference	User	32		false
Created by	String	(empty)	40		false
Device Details	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Created	Date/Time	(empty)	40		false
Customer Document	String	(empty)	40		false
Work Status	String	(empty)	40		false
Assignment Group	Reference	Group	32		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false

Database Number – Auto Number Configuration

The Database Number field should be configured as an automatically generated identifier where required by the lab instructions.

Configure:

- **Auto Number:** Enabled
- **Prefix:** Use the prefix specified by your project/lab
- **Number format:** Use the required number format

This ensures that each record can be uniquely identified.

Assignment to – Reference Field

The **Assignment to** field should be configured as a **Reference** field pointing to the **User** table.

This allows the record to be associated with an existing ServiceNow user instead of storing the user's name as unrestricted text.

Form Design

After creating the fields:

1. Open the table.
2. Select **Form Design**.
3. Arrange the fields in the required order.
4. Place related fields together logically.
5. Save the form.
6. Open a sample **Network Database** record to verify the fields.

The screenshot shows the ServiceNow interface for configuring the 'Network Database' table. The 'Columns' tab is active, displaying a list of dictionary entries. The entries are as follows:

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40	40	false
Created by	String	(empty)	40	40	false
Sys ID	Sys ID (GUID)	(empty)	32	40	false
Created	Date/Time	(empty)	40	40	false
Updated by	String	(empty)	40	40	false
Updates	Integer	(empty)	40	40	false

Outcome

The data architecture establishes a structured foundation for the Automated Network Request Management solution.

After completion:

- A dedicated custom table exists.
- Required network/request fields are available.
- Records can be uniquely identified.

- Users can be referenced through ServiceNow user records.
- Request information is stored in a consistent structure.
- The data can subsequently be consumed by approval flows, fulfilment automation, notifications, dashboards, and reporting.

Replicability Note

Note: The exact field types and naming should follow the capstone/lab specification. In particular, fields such as **Total Amount**, **Mobile Number**, **Approval Status**, and **Type of Connection** are currently specified as String fields in the supplied requirements. If later phases use these values for calculations, filtering, or conditional automation, we should review whether a more specialized ServiceNow field type is appropriate.